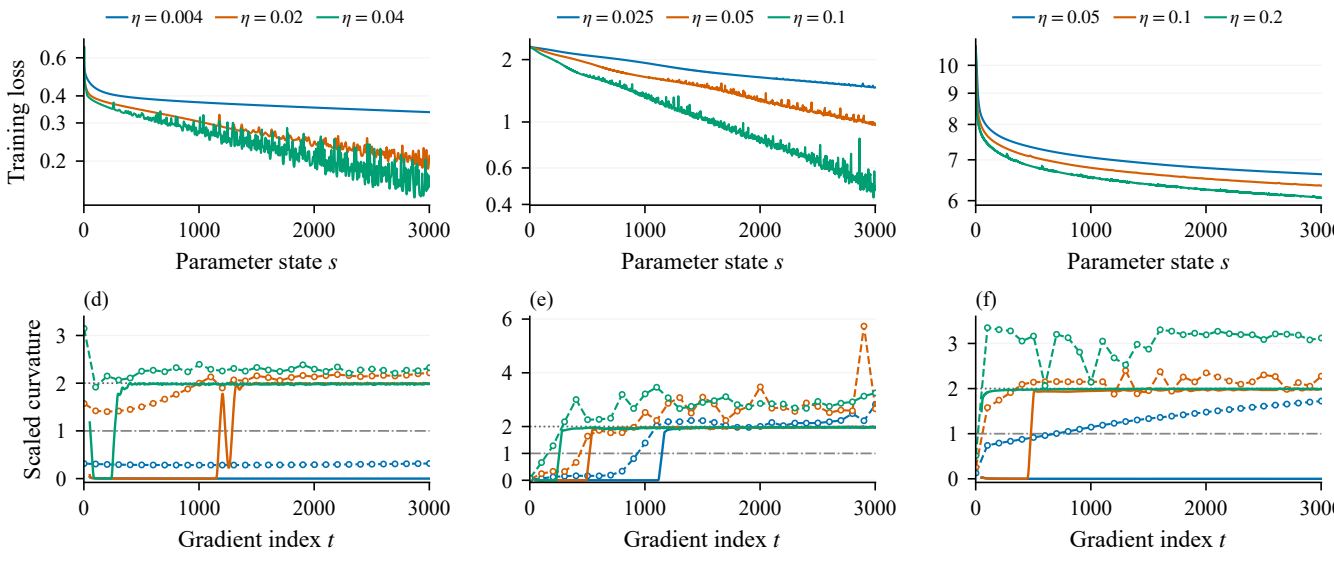

(a) CIFAR-10 / MLP
(b) CIFAR-10 / ResNet-32
(c) WikiText-2 / Transformer
η = 0.004
η = 0.02
η = 0.04
η = 0.025
η = 0.05
η = 0.1
η = 0.05
η = 0.1
η = 0.2
Training loss
Parameter state s
(d)
(e)
(f)
Scaled curvature
Gradient index t
ηκ̂t (50-pair mean)
ηλmax
1: reversal
2: quadratic loss threshold